INCH-POUND

MIL-C-70532 (AR)
AMENDMENT 9
26 August 2002
SUPERSEDING
AMENDMENT 8
18 December 1998

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE, RA, XM913 AND XM927 (WITH WARHEAD AND MOTOR BODY) LOADING, ASSEMBLING AND PACKING

MIL-C-70532(AR) was inactivated after 21 August 1998 for new design.

This amendment forms a part of MIL-C-70532 (AR), dated 29 October 1990, and is approved for use by the U.S. Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Change "XM913" to "M913" and "XM927" to "M927" and wherever they appear.

* 2.1.1, Delete "MIL-STD-453" in its entirety and substitute the following:

"ASTM E 1742 - Standard Practice for Radiographic Examination"

PAGE 2

2.1.2: Drawings:

Change "(See 6.9)" to "(See 6.11)"

Change "9290990" to "9390990"

Dwg. 12624495, add: "PA111," after "assembly"

Dwg. 12624496, change: "Ammunition" to "Ammo"

Dwg. 12934903, change: "XM912" to "M913"

Dwg. 12630699, delete in its entirety.

* Add new drawing 12926808 as follows: "12926808 - Projectile, HE, RA, 105MM, M927"

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Drawing 19-48-4231/45-20PM1006, Add: "and" after "Ammunitions" and delete "Cyclindrical" and substitute "Cylindrical"

PAGE 3

3.1.1, Line 2, Change "operation" to "Operations"

3.2.3, Change "grains" to "grain" two places

PAGE 4

3.2.3, Delete in its entirety.

3.2.4, Delete in its entirety.

PAGE 5

3.3.5, Change "thousands" to "thousandths"

PAGE 6

Delete Figure 1 and substitute the following new Figure included.

PAGE 7

3.6, Capitalize "proving" in title.

* 3.6.1.2.1, Add at end:

"The short range of projectiles caused by failure of the delay assembly to ignite the rocket motor shall not be included in the RPE calculation."

PAGE 8

Delete 3.6.1.2.3 in its entirety and substitute:

"3.6.1.2.3 M913 cartridge muzzle velocity. The average muzzle velocity for cartridges fired at 70 degree F shall be not less than 643 meters per second (mps) nor greater than 659 mps and the standard deviation shall not exceed 2.44 mps."

3.6.2.1, Change title from "Pressure" to "M913 cartridge pressure"

Add 3.6.2.2 as follows:

3.6.2.2 M927 cartridge pressure. The pressure of any individual cartridge, fires at +145 degree F, shall not produce a PIMP greater than 44,800 psi. "

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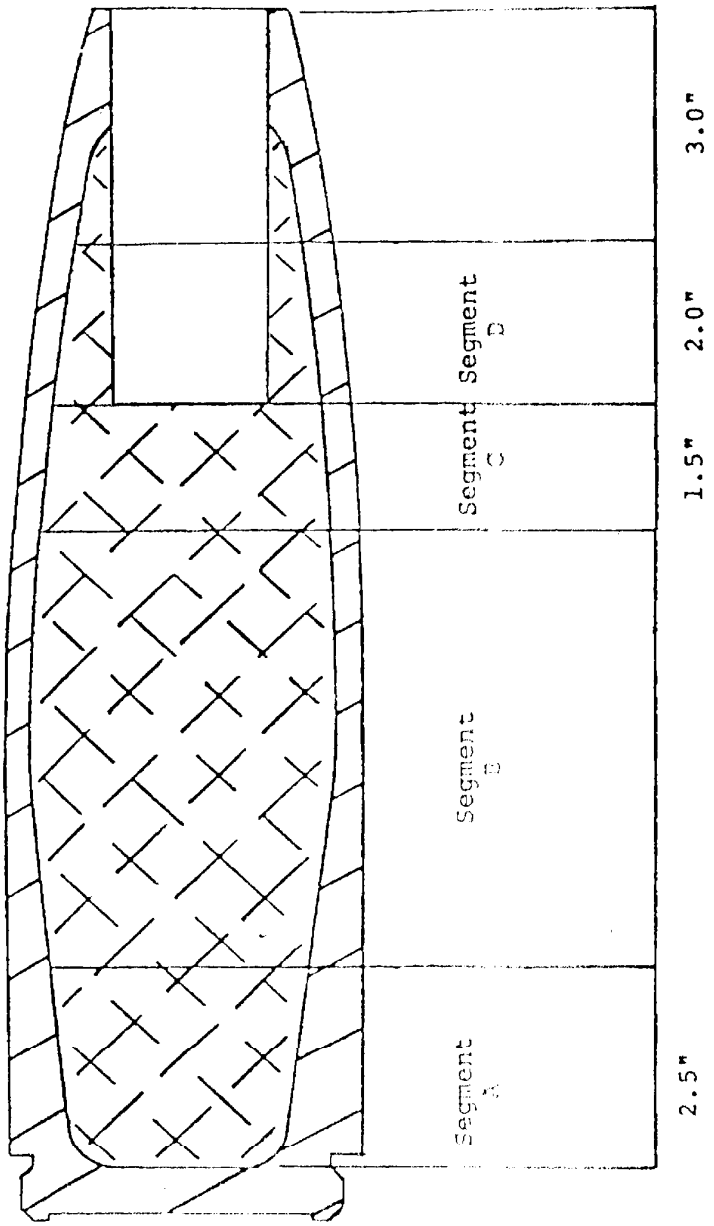


Figure 1. Cartridge, 105MM, IIERA, M913/M927 Warhead Segments.

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PAGE 9

3.10.5 Delete in its entirety.

Add 3.11 as follows:

"3.11 Delay shear seal push-out force. The force required to push a delay out of a rocket motor after pressing of the shear seal shall be 950 pounds to 1300 pounds."

PAGE 10

4.3.1, Line 11, delete: "Flash reducer, 12630698, 10" in its entirety.

* 4.3.1, Under "Projectile", add "/12926808" after "9381147"

Table I, Lines 10, 11 and 12, delete "Flash reducer, (dwg. 12630698), Examinations for defects, 10, 3.2, 4.4.2.18" in its entirety.

Table I, Change the paragraph reference in note a. from "4.4.2.1" to "4.4.2.2"

PAGE 13

4.4.1f: Delete "up tp four (4) consecutively produced lots of"

4.4.1h: Delete in its entirety and substitute the following:

"h. Delay assemblies from one manufacturer, with constituent chemicals from one lot each, and having the same fuel/oxidizer ratio."

4.4.1k: Delete "Potassium sulfate from not more than one lot." in its entirety.

4.4.1k: Delete j in its entirety and substitute: "j. Shear seals from not more than one lot interfix from one manufacturer."

4.4.1.1, Delete "M30 propellant from not more than one lot."

PAGES 15 THRU 18

4.4.2.1 – 4.4.2.4, Next Higher Assembly, Add: "/12926808"

* 4.4.2.3, Delete Major defect 103 in its entirety and substitute the following:

"103. Inside diameter of liner after swaging (at threaded area) min. III Gage"

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PAGES 19 THRU 25

4.4.2.5 – 4.4.2.7, 4.4.2.9 – 4.4.2.11, Drawing Number, add : "/12926808"

4.4.2.8, Delete in its entirety.

4.4.2.9, Major 103, delete "3.2.2" and substitute "3.2.3"

4.4.2.11, Add Special 1 as follows:

"1. Weight zone markings, missing or incorrect 100% 3.2 Scale/Visual"

4.4.2.11, Major 107, change from "Test/4.4.3.6/4.5.2" to "Visual/(Witness during assembly) /4.4.3.6"

PAGE 27

4.4.2.13:

Drawing Number, delete "9390990/9391033" and substitute "12938083/8595386"

Next Higher Assembly, add: "9390990/9391033"

PAGE 28

4.4.2.14, Drawing Number, delete "9205472/9382042" and substitute "12630699/9205472"

4.4.2.14, Critical defect 1, Add: "/Scale" after "Gage"

PAGE 29

4.4.2.15:

Drawing Number, delete "9391033" and substitute "9390990/9391033"

Major 101, delete "Incorrect sequence or increments (M67)" and substitute "Incorrect sequence of increments (M67)"

Major 102, delete "Foil side of bag, charge 5, not facing primer (M67)" and substitute "Foil side of bag, charge 5, not facing primer (M67), if applicable."

PAGE 31

4.4.2.17 Majors 101, 102, 103, Add: "/SMTE" after "Gage"

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Add Major defect 104 and 105 as follows :

"104 Evidence of bumps, blisters, pits, splits, delaminations, corrosion, or other defects immediately prior to assembly into motor body 100% 3.1 Visual

105 Failure to meet push-out force requirements 4.4.3.8.3.11 Test"

PAGES 32 THRU 34

4.4.2.18 – 4.4.2.20, Delete in their entirety.

PAGE 35

4.4.2.21, Delete "12630699" and substitute "12624496"

PAGE 37

4.4.2.23, Delete "12934900" and substitute "12624496"

Add "12934900" to "Next Higher Assembly"

Delete Title and substitute: "Packing and Marking of Sealed Fiber Container and Metal Container Components (prior to sealing of Metal Container)"

PAGE 38

4.4.2.24:

Next Higher Assembly, Delete drawing numbers in their entirety and substitute the following: "12929177/12929178"

Add the following new major defects as follows:

"106	Insufficient fillers for tight pack	Level III	3.1	Visual
107	Filler size (min.)	Level III	3.1	Gage
108	Filler not present underneath cloth strap	Level III	3.1	Visual"

Minor defect 202, delete in its entirety and substitute the following:

"202 Evidence of poor workmanship Level V 3.10 Visual"

PAGE 41

4.4.3.1.2, Line 6, delete "6.10" and substitute "6.13"

4.4.3.3, Line 3 thru 8, delete and substitute the following:

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"Each assembly shall be subjected to a radiographic examination in a plane that will clearly define all of the Motor Body components."

PAGE 42

4.4.3.4:

Line 2, delete "4.5.3" and substitute "4.5.7"

Add Critical defect 103 as follows:

"103 Pressure in excess of 44,800 psi (See 3.6.2.2)"

Special a, delete "6.15" and substitute "6.14"

* Major 102, delete in its entirety and substitute the following:

"102 Rocket motor fails to ignite after delay assembly expulsion"

Major defect 103, delete "Muzzle velocity" and substitute "M913 cartridge muzzle velocity"

* Add new Major 104 as follows:

"104 Delay assembly fails to ignite or expel"

4.4.3.4.1, Change "4.5.8.1" to "4.5.7.1"

* 4.4.3.4.1.1, Delete last sentence in its entirety and substitute the following:

"If any critical defect or major defect other than 104 occurs, the lot shall be rejected"

PAGE 43

* 4.4.3.4.1.2, Delete second sentence in its entirety and substitute the following:

"If any critical defect or major defect other than 104 occurs, the lot shall be rejected"

4.4.3.4.2.3, Line 6, delete "6.13" and substitute "6.12"

* 4.4.3.4.2.3, Delete in its entirety and substitute the following:

4.4.3.4.2.3 Temperature test failures. If major defect 102 occurs during +145 degree F safety testing, the lot shall be rejected. If major defect 102 occurs during –50 degree F safety testing, the lot shall be referred to the procuring agency for disposition (see 6.12)."

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* Add new paragraph 4.4.3.4.4 as follows:

"4.4.3.4.4 Failure of delay assembly. If major defect 104 occurs a total of two or more times in the proving ground test (all phases), the lot shall be rejected. If major defect 104 occurs once in the proving ground test an additional test for all phases shall be conducted. The lot shall be rejected if any critical or major defect occurs."

4.4.3.5, Second sentence, add a "," after the word "change"

PAGE 44

4.4.3.7, Line 12, delete "6.11" and substitute "6.16"

Add new paragraph 4.4.3.8 as follows:

"4.4.3.8 Delay shear seal push-out force qualification test. Prior to assembly of each lot of shear seals into each manufacturer's lot interfix of rocket motor bodies, five sample assemblies shall be pressed with the load, dwell, and press tooling to be used in production. Each sample assembly shall be radiographically inspected in accordance with 4.4.3.3. Each sample delay shall be pressed out, and the force required shall be recorded. If any sample falls outside the allowable range, the motor body, seal, load and dwell combination shall not be used in production. Rate of application of test load shall be .5 inches per minute."

PAGE 48

4.5.2, Line 5, delete "rocket selector cap and the"

4.5.4, Change "4.4.3.6" to "4.4.3.5"

4.5.5, Delete in its entirety

PAGE 49

4.5.6, 4.5.6.1, 4.5.6.1.1, 4.5.6.1.2, Delete in their entirety.

4.5.7, Change "M557" to M739 series"

4.5.7b, Add the following:

"6. Chamber pressure (+145 degree firing)
7. Range and deflection of impacts.
8. Rocket motor ignition delay time."

PAGE 50

4.5.7c, Delete in its entirety.

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4.5.7.1, Delete in its entirety and substitute the following :

"4.5.7.1 Functioning/precision/muzzle velocity. Cartridges for this test shall be conditioned at 70 ± 2 ½ degree F for 16 hours (minimum) prior to testing. The M913 test cartridges shall have their muzzle velocity corrected for projectile weight and gun tube wear in accordance with the acceptable M913 firing tables. At least three warmer/spotter cartridges shall be fired prior to firing test cartridges. Time lapses of more than 15 minutes between consecutive firings shall not be permitted without firing additional warmer cartridges. The test shall be fired at 700 to 800 MILS QE. Range probable error shall be calculated for the test cartridges.

Two master calibration cartridges shall be fired with each 10, M913 test cartridges with muzzle velocity and chamber pressure recorded for information only."

PAGE 51

4.5.7.2, Delete in its entirety and substitute the following:

4.5.7.2 Safety. Cartridges for this test shall be divided into two equal groups and temperature conditioned for 16 hours (minimum) prior to testing. Group 1 shall be conditioned and fired at – 50 + 5 degrees F. Group 2 shall be conditioned and fired at +145 – 5 degrees F. The two groups shall be fired at a convenient elevation selected by the proving ground."

5.4.1, Change "19048-4231" to "19-48-4231"

PAGE 52

6.3, Line 4, Delete "AMSMC-QAR-I (D)" and substitute "AMSTA-AR-QAA-R"

PAGE 53

6.7, Line 23, Delete "4.3.2" and substitute "4.5.7.2"

6.8, Line 5, Delete "AMSMC-QAR-Q" and substitute "AMSTA-AR-QAA-R"

PAGE 54

6.9.1 Delete in its entirety and substitute the following:

"6.9.1 Radiographic film sources. Approved list of film types and vendors for use in radiographing this projectile:

Eastman Kodak Company
1187 Ridge Road, West
Rochester, NY 14650
Film Type: AA, T, AX-1, M, MX

Agfa Corporation
NDT Systems
100 Challanger Road
Ridgefiled Park, NJ 07660
Film Type: Agfa D7

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Fuji Industrial Radiographic Products
Corporate Headquarters
20 North Ave.
Larchmont, NY 10538
Film Type : Fuji 100"

6.10 Delete in its entirety and substitute the following:

"6.10 Ballistic test projectiles. All projectile tested by a Government Proving Ground for Ballistic Acceptance Test shall have three markings 120 degrees apart, painted at the LAP manufacturing facility across the warhead/motor body joint. See Figure 3."

Add new Figure 3 as included page 54a.

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Figure 3.

Ballistic Test Projectile Marking

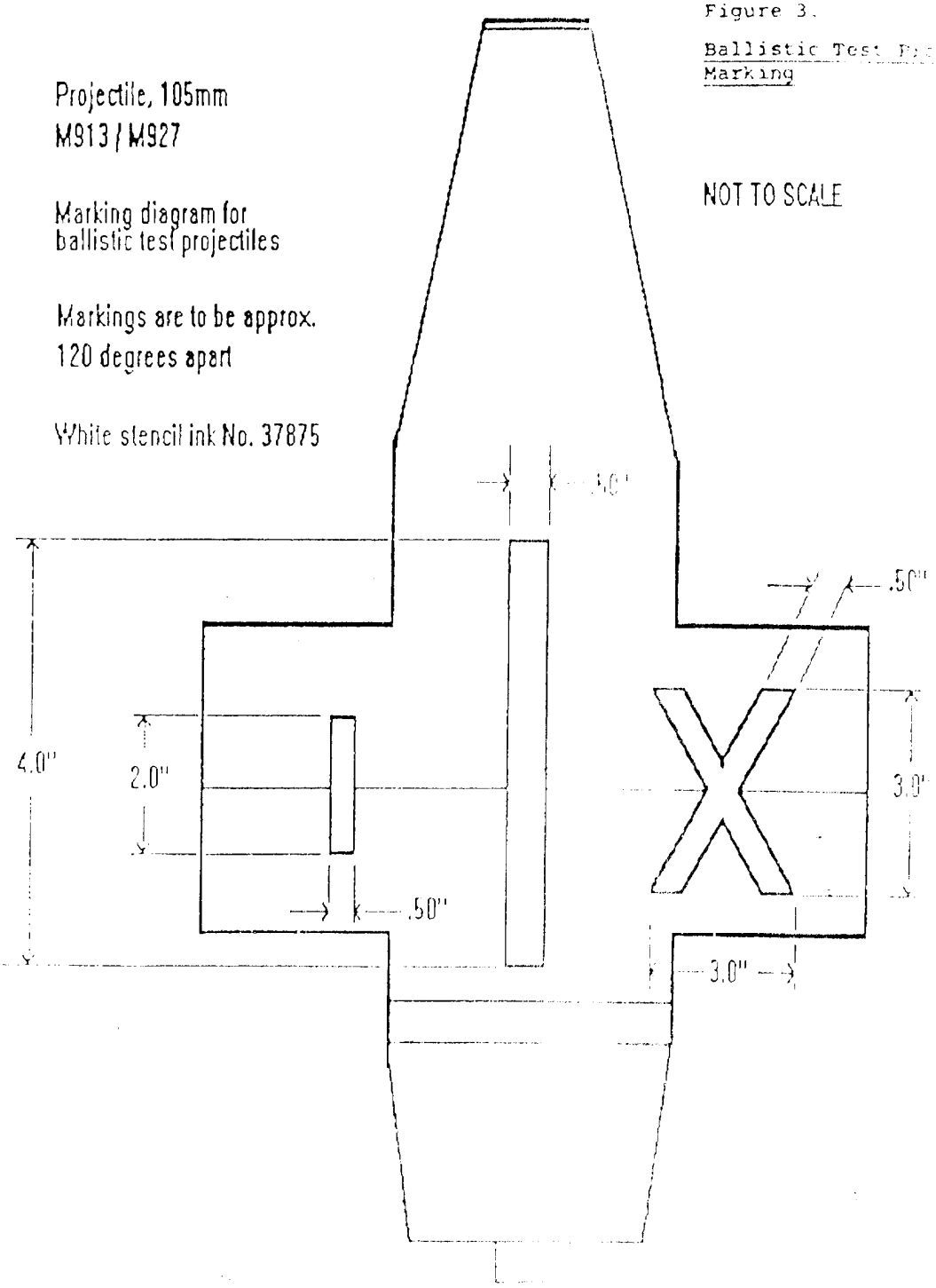
Projectile, 105mm
M913 / M927

Marking diagram for
ballistic test projectiles

Markings are to be approx.
120 degrees apart

White stencil ink No. 37875

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